

## Multiple Choice (10 marks)

1. In Python 3, what is `['a', 'Chemistry', 0, 1][1]`?
  - (a) a
  - (b) Chemistry
  - (c) 0
  - (d) 1
2. Which is the smallest asymptotically?
  - (a)  $O(1)$
  - (b)  $O(n)$
  - (c)  $O(n^2)$
  - (d)  $O(\log n)$
3. Let A be a finite set with m elements, and let B be a finite set with n elements. The number of distinct functions mapping A into B is
  - (a)  $n^m$
  - (b)  $n!/(n - m)!$
  - (c)  $n!$
  - (d)  $n!/(m!(n - m)!)$
4. In Python 3, what is the output of `['Hi!'] * 4`?  
`'Hi!', 'Hi!', 'Hi!', 'Hi!'`  
`'Hi!' * 4`
  - (a) Error
  - (b) None of the above.
5. Let  $f(X) = \text{if } x = 1 \text{ then } 0 \text{ else } [x * f(x - 1) + x**2]$ . The value of  $f(4)$  is
  - (a) 53
  - (b) 29
  - (c) 50
  - (d) 100

## True / False (10 marks)

*Answer True or False*

6. True or False: In Python 3, the expression `['a', 'Chemistry', 0, 1][-3]` evaluates to `'Chemistry'`.
7. True or False: All bit strings with an even number of zeros can be described by a regular expression.
8. True or False: Finding a shortest cycle in an undirected graph  $G$  can be solved in polynomial time.
9. True or False: In Python 3, the function `max([1, 2, 3, 4])` returns the value 4 as the maximum element of the list.
10. True or False: Merge sort has a running time that is least affected by the initial ordering of the input compared to other sorting algorithms.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function to extract all the adjacent coordinates of the given coordinate tuple.
12. Write a python function to check whether the given array is monotonic or not.

*End of Paper*